Part D: Naming & Formula of Polyatomic Ionic Compounds

April 10, 2019 3:16 PM

PART D: NAMING & FORMULA OF POLYATOMIC IONIC COMPOUNDS

REMEMBER THAT:

A **polyatomic** ion is composed of more than one type of atom joined by covalent bonds.

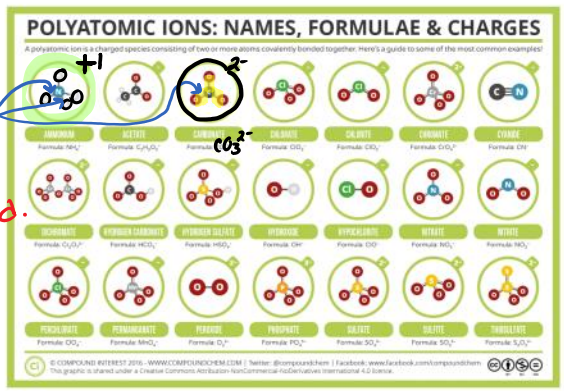
Because polyatomic ions carry an electric charge, they cannot exist on their own. ↳ part of an ionic compound.

An example of a polyatomic ion is:

carbonate, CO_3^{-2}

The figure *right* (and your data booklet) lists some common polyatomic ions.

You do not have to memorize the names.



PRACTICE Write the name or formula of the following polyatomic ions:

1. NH_4^+ — ammonium ion
2. CrO_4^{-2} — chromate ion
3. Acetate — CH_3COO^-
4. CN^- — cyanide ion
5. SO_4^{-2} — sulphate
6. Sulfite — SO_3^{2-}
7. Hypochlorite — ClO^-
8. OH^- — hydroxide ion
9. Bisulphite — HSO_3^-
10. HCO_3^- — hydrogen carbonate (or bicarbonate)

* when we name with polyatomics we NEVER change name in any way.

- Polyatomic ions are treated just like other ion but when naming compounds we DO NOT change their endings (they are already ions)
  - Example: NO_3^{-1} is nitrate, not nitratide ~~no "ide" ending!~~
- You will recognize that there is a polyatomic ion present, because the compound will look like it has more than TWO Parts. It doesn't! Just split up the compound after the metal, and then find the other ion in your data booklet. most polyatomics have all non-metals.
- Remember to use BRACKETS in the formula if more than one is present, as you will see from examples.

2 3

1. FROM CHEMICAL FORMULA --> COMPOUND NAME

$CaCO_3$ (3)

Step 1: How do you know you are dealing with a polyatomic ion?

there are more than 2 elements

Step 2: don't change ending

Step 1: How do you know you are dealing with a polyatomic ion?
There are more than 2 elements

Step 2: identify metal: Ca^{2+} = calcium ion

Step 3: identify non-metal part: CO_3^{2-} = carbonate ion

don't change ending of polyatomic ion

Name: Calcium carbonate

NH_4Cl

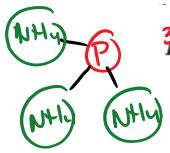
Step 1: 3 elements are involved: N, H, Cl => has polyatomic ions.

Step 2: NH_4^+ is a metal polyatomic ion (+) = Ammonium

Step 3: Cl^- is a non-metal ion = chloride

Name: ammonium chloride

PRACTICE Write the name of the following compounds that contain ***polyatomic ions***.



3xLi

periodic table

polyatomic ion table

Example: Write the name of the following compounds.

1. Li_3PO_4 — Li^{+1} PO_4^{-3} — Li_3PO_4 lithium phosphate
2. $(NH_4)_3P$ — NH_4^{+1} P^{-3} — $(NH_4)_3P$ ammonium phosphide
3. $Ca(OH)_2$ — Ca^{+2} OH^{-1} — $Ca(OH)_2$ calcium hydroxide
4. $Fe(OH)_3$ — Fe^{+3} OH^{-1} — $Fe(OH)_3$ iron (III) hydroxide.
5. $Mn(SO_4)_2$ — Mn^{+4} SO_4^{-2} (SO_4^{-2}) — $Mn(SO_4)_2$ manganese (IV) sulphate.

2. FROM COMPOUND NAME --> CHEMICAL FORMULA

Aluminum chromate

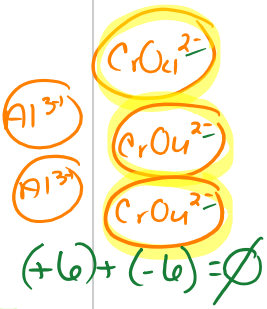
metal ion: Al^{3+} — non-metal (polyatomic ion): CrO_4^{2-}

SWAP! Drop! 2 : 3

$Al_2(CrO_4)_3$

Note that if there is more than one of the polyatomic compounds, you must put it in parenthesis and add subscripts.

Cross the charges circled to balance and neutralize the compound.



Notice the use of brackets in the formula to allow the ratio of ions to be shown correctly.

Brackets are dropped if the ion is **not polyatomic** or if the **ratio** number outside the brackets is 1

When you read the formula, you should always remember that the *ratio numbers and brackets are implied.*

FOR EXAMPLE: Na_2CO_3 as sodium carbonate with 2 Na^+ ions for every 1 CO_3^{-2} ion.

$Ca(OH)_2$ as calcium hydroxide with 1 Ca^{2+} ions for every 2 OH^- ion.

Magnesium sulfate ← polyatomic

Mg^{2+} SO_4^{2-}

Mg^{2+} SO_4^{2-}

2 : 2 simplify ÷2

1 : 1

$Mg_{\square}(SO_4)_{\square}$ ⇒ $MgSO_4$

drop brackets

PRACTICE

1. Write the formula of each of the following compounds:

Ba^{2+} NO_3^{-} → 1 : 2

a. barium nitrate $Ba(NO_3)_2$

b. potassium carbonate

c. nickel (II) sulfate

d. magnesium phosphate

e. sodium dichromate

a. barium nitrate $Ba(NO_3)_2$

b. potassium carbonate K_2CO_3

c. nickel (II) sulfate $NiSO_4$

d. magnesium phosphate $Mg_3(PO_4)_2$

e. sodium dichromate $Na_2Cr_2O_7$

f. iron (II) chromate $FeCrO_4$

g. lead (IV) acetate $Pb(CH_3COO)_4$

h. ammonium sulfate $(NH_4)_2SO_4$

Homework

ASSIGNMENT #6: Ionic Compounds with POLYATOMIC IONS Naming & Formula Review + MAZE on page 27

This assignment is to be completed below in the space provided.

Part 1 – Write the name for each of the following ionic compounds.

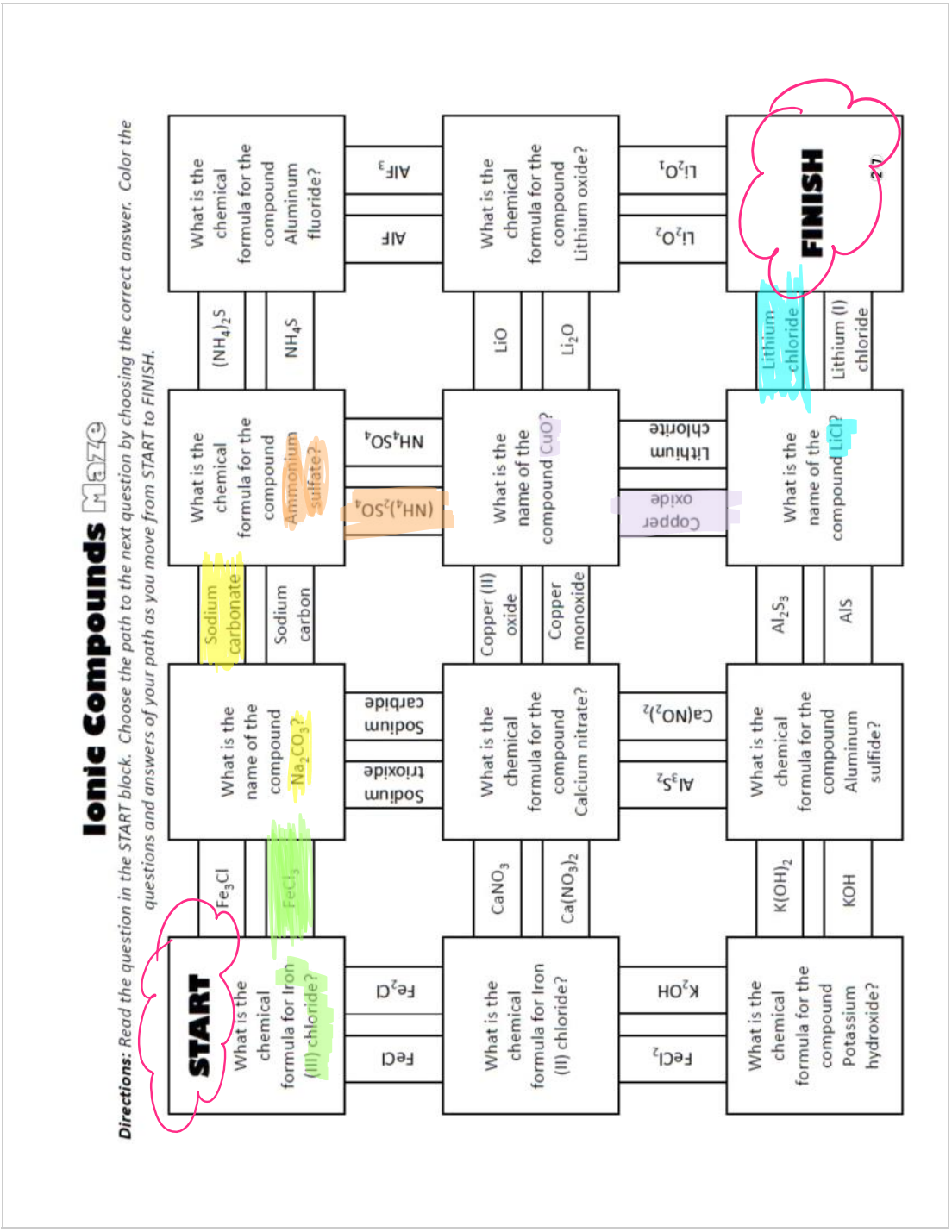
1. Na_2CO_3 sodium carbonate
2. $Fe(OH)_3$ _____
3. KCH_3COO _____
4. $Co(ClO)_2$ _____
5. $(NH_4)_3PO_4$ _____
6. $Mg_3(PO_4)_2$ _____
7. $Ca(CH_3COO)_2$ _____
8. $Mg_3(PO_4)_2$ _____
9. $(NH_4)_3P$ ammonium phosphide
10. $Ni(HS)_3$ nickel (III) bisulphide
11. $(NH_4)_3PO_4$ _____
12. $CuCN$ copper(I) cyanide
13. $Al(OH)_3$ aluminium hydroxide
14. $Mn(SO_3)_2$ manganese(II) sulphite.

HW April 16

Part 2 – Write the formula for each of the following ionic compounds:

Compound Name	Work	Formula
15. calcium carbonate		$CaCO_3$
16. manganese (III) chlorate	Mn^{3+} ClO_3^-	$Mn(ClO_3)_3$
17. lithium fluoride	Li^+ F^-	LiF
18. potassium permanganate	K^+ MnO_4^-	$KMnO_4$
19. sodium chromate	Na^+ CrO_4^{2-}	Na_2CrO_4
20. ammonium nitrate	NH_4^+ NO_3^-	NH_4NO_3
21. lithium hydroxide	Li^+ OH^-	$LiOH$
22. aluminum hydroxide	Al^{3+} OH^-	$Al(OH)_3$
23. lead (II) perchlorate	Pb^{2+} ClO_4^-	$Pb(ClO_4)_2$
24. iron (III) hydrogen sulfide	Fe^{3+} HS^-	$Fe(HS)_3$
25. vanadium (V) nitrate	V^{5+} NO_3^-	$V(NO_3)_5$
26. chromium (II) nitrite	Cr^{2+} NO_2^-	$Cr(NO_2)_2$
27. nickel (III) sulfite	Ni^{3+} SO_3^{2-}	Ni_2SO_3

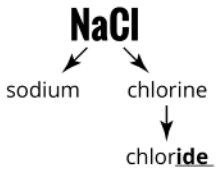
26


Ionic Compounds Maze
Directions: Read the question in the START block. Choose the path to the next question by choosing the correct answer. Color the questions and answers of your path as you move from START to FINISH.
START
What is the chemical formula for Iron (III) chloride?
FeCl
Fe_2Cl
Fe_3Cl
$FeCl_3$
What is the name of the compound Na_2CO_3?
Sodium trioxide carbide
Sodium carbide
Sodium carbon
Sodium carbonate
What is the chemical formula for the compound Ammonium sulfate?
NH_4SO_4
$(NH_4)_2SO_4$
NH_4S
$(NH_4)_2S$
What is the chemical formula for the compound Aluminum fluoride?
AlF
AlF_3
What is the chemical formula for Iron (II) chloride?
$FeCl_2$
Fe_2Cl
$CaNO_3$
$Ca(NO_3)_2$
What is the chemical formula for the compound Calcium nitrate?
$Ca(NO_2)_2$
Al_3S_2
Copper (II) oxide
Copper monoxide
What is the name of the compound CuO?
Copper oxide
Lithium chlorite
LiO
Li_2O
What is the chemical formula for the compound Lithium oxide?
Li_2O_1
Li_2O_2
What is the chemical formula for the compound Potassium hydroxide?
$FeCl_2$
K_2OH
$K(OH)_2$
KOH
What is the chemical formula for the compound Aluminum sulfide?
Al_2S_3
AlS
What is the name of the compound LiCl?
Lithium chloride
Lithium (I) chloride
FINISH

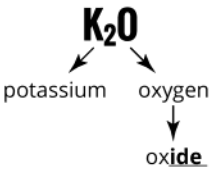
Writing Compound Names

Ionic Bonds
(a bond between a metal and a nonmetal)

Naming a Binary Ionic Compound
(two elements with no transition metals)

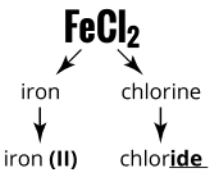


sodium chloride

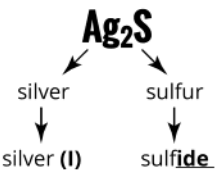


potassium oxide

Naming a Compound with a Transition Metal



iron (II) chloride

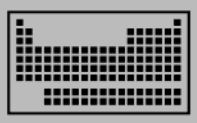


silver (I) sulfide

you can figure out this number based on the number of atoms of the second element

Element or Polyatomic Ion?

Elements are found on the **periodic table.**



Elements look like this:

H K O OR Cu Ag

one capital letter

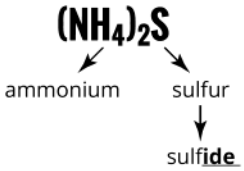
one capital letter and one lowercase letter

Polyatomic ions are groups of two or more elements.

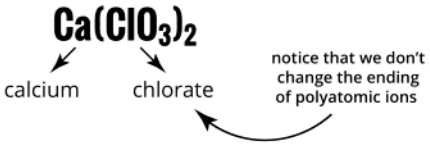
ClO_3 NH_4 OH

They stick together.

Naming a Compound with a Polyatomic Ion



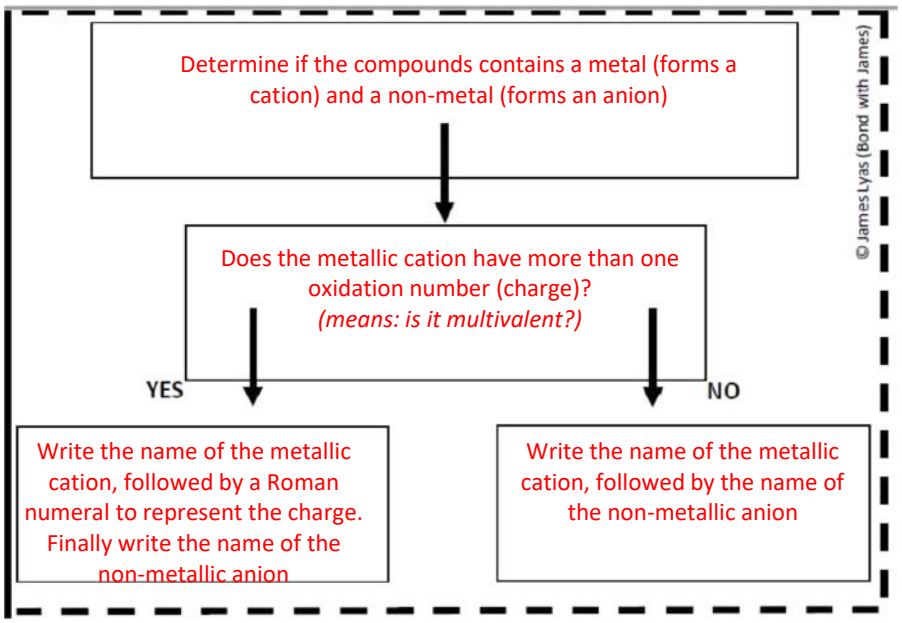
ammonium sulfide



calcium chlorate

28

Steps for Naming Binary Ionic Compounds



Directions: Cut out the organizer and smaller pieces along the dotted lines only. Then tape or glue the pieces to help construct a logical sequence of steps for naming ionic compounds.

Recreating, scanning, electronic copying, file sharing, redistributing, editing, selling, or posting this item (or any part thereof) on the Internet, or with others, is strictly prohibited. Violations are subject to the penalties of the Digital Millennium Copyright Act. © James Lyas (Bond with James)

Determine if the compound contains a metal (forms a cation) and a non-metal (forms an anion).

Write the name of the metallic cation, followed by a Roman numeral to represent the charge. Finally write the name of the non-metallic anion.

Write the name of the metallic cation, followed by the name of the non-metallic anion.

Does the metallic cation have more than one oxidation number?

29

30